

200 AI Engineering Interview Questions

with Answers — 2025 Edition

GenAI · LLMs · RAG · Agents · MCP · Fine-Tuning · ML · DL

Q1–20 · LLM Fundamentals

Q21–40 · Prompt Engineering

Q41–60 · RAG & Retrieval

Q61–80 · AI Agents

Q81–100 · Training & Fine-Tuning

Q101–120 · Inference & Serving

Q121–140 · AI Automation

Q141–160 · Machine Learning

Q161–180 · Deep Learning

Q181–200 · Career & Products

Difficulty Levels:

EASY

MEDIUM

HARD

How to Use This Guide

- EASY questions test foundational knowledge — expected of all candidates at all levels.
- MEDIUM questions test design depth and trade-off understanding — show your reasoning.
- HARD questions cover system design, theory, and production challenges — senior/staff level.
- Use answers as revision anchors — expand with your own examples and real-world experiences.

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1 What is a Large Language Model (LLM)?

Answer:

- Neural network trained on massive text to predict the next token.
- Uses Transformer architecture with self-attention at its core.
- Can perform many tasks zero-shot — no task-specific training needed.

2 What is tokenization and why does it matter?

Answer:

- Converts raw text into tokens (subwords/words) the model processes.
- Affects vocabulary size, sequence length, cost & multilingual coverage.
- BPE and SentencePiece are the dominant algorithms in production.

3 What is temperature in LLM generation?

Answer:

- Scales logits before softmax: low T = deterministic, high T = random.
- T=0 is greedy decoding; T~0.7 balances creativity and coherence.
- Controls diversity of outputs without changing model weights.

4 What is the difference between zero-shot, few-shot prompting?

Answer:

- Zero-shot: task described in words only, no examples provided.
- Few-shot: a handful of examples guide the format and reasoning style.
- More examples generally improve accuracy on structured tasks.

5 What is hallucination in LLMs and why does it occur?

Answer:

- Model generates plausible-sounding but factually incorrect content.
- Caused by knowledge gaps, distribution shift, overconfident prediction.
- Mitigated via RAG, grounding, fact-checking layers, and RLHF.

6 What is prompt engineering?

Answer:

- Crafting inputs to reliably elicit desired outputs from an LLM.
- Techniques: zero-shot, few-shot, chain-of-thought, persona setting.
- Critical skill — better prompts beat bigger models in many tasks.

7 What is the context window of an LLM?

Answer:

- Max tokens (input + output) processed in one forward pass.
- Limited by $O(n^2)$ memory cost of attention; larger = more expensive.
- Claude supports up to 200k tokens; practical limit depends on cost.

8 What is an embedding and what are they used for?

Answer:

- Dense real-valued vector representing text in a semantic space.
- Similar meanings map to nearby vectors — powers semantic search.

11 What is RAG (Retrieval-Augmented Generation)?

Answer:

- Combines a retriever (searches knowledge base) with an LLM generator.
- Retrieved chunks injected into prompt — grounds response in real data.
- No retraining needed; keeps knowledge fresh and up-to-date.

12 What is fine-tuning in LLMs?

Answer:

- Continuing training of a pretrained model on a smaller domain dataset.
- Types: full fine-tuning, PEFT (LoRA, adapters), instruction tuning.
- Cheaper than pretraining but risks catastrophic forgetting.

13 What is the attention mechanism in a Transformer?

Answer:

- $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$ — weighted sum of values.
- Allows every token to attend to every other token in $O(n^2)$ time.
- Multi-head attention runs h parallel heads, each learning different patterns.

14 What is greedy decoding vs beam search?

Answer:

- Greedy: pick highest-probability token each step — fast but suboptimal.
- Beam search: keep top-B sequences at each step — explores more paths.
- Beam is $B \times$ more compute; still not globally optimal but better quality.

15 What is top-k and top-p (nucleus) sampling?

Answer:

- Top-k: sample only from k highest-probability tokens at each step.
- Top-p: sample from smallest set with cumulative probability $\geq p$.
- Top-p adapts the candidate pool dynamically — often more coherent.

16 What is autoregressive generation?

Answer:

- Tokens generated one at a time, each conditioned on all previous tokens.
- $P(x_1 \dots x_n) = \prod P(x_i | x_{1:n-1})$ — inherently sequential process.
- Parallelising output tokens is non-trivial (see speculative decoding).

17 What is a system prompt?

Answer:

- Hidden prepended prompt setting the model's persona and constraints.
- Shapes tone, safety boundaries, and task focus across a conversation.
- Operators use system prompts to customize Claude for their product.

What is chain-of-thought (CoT) prompting?

21

Answer:

- Adds intermediate reasoning steps — 'Let's think step by step.'
- Improves accuracy on multi-step arithmetic, logic, and commonsense tasks.
- Zero-shot CoT uses the phrase itself; few-shot CoT provides examples.

What is self-consistency prompting?

22

Answer:

- Sample N diverse reasoning chains, take majority-vote final answer.
- Errors in individual chains are uncorrelated — voting cancels them.
- Effective but N× inference cost; 5-10 samples usually sufficient.

What is ReAct (Reasoning + Acting) prompting?

23

Answer:

- Interleaves reasoning traces with action calls (search, code exec, APIs).
- Model thinks → acts → observes result → thinks again in a loop.
- Foundation of modern AI agents that interact with external tools.

What is prompt injection and how is it mitigated?

24

Answer:

- Attack where malicious text in retrieved data overrides system instructions.
- Mitigations: input sanitization, privilege separation, output filtering.
- Fine-tuning models to be robust to injection patterns is a key defense.

What is few-shot learning vs in-context learning?

25

Answer:

- Few-shot: examples placed in the context window guide the model.
- In-context learning: model adapts to a task from examples without gradient updates.
- LLMs perform remarkable in-context learning at scale — emergent ability.

What is meta-prompting?

26

Answer:

- Using an LLM to generate or refine prompts for another LLM task.
- Automates the prompt engineering process — useful for complex tasks.
- Reduces manual iteration; models can self-improve their own prompts.

What is role prompting?

27

Answer:

- Assigning a persona to the model: 'You are a senior software engineer.'
- Guides tone, knowledge depth, and problem-solving style significantly.
- Effective especially for domain-specific tasks like legal or medical Q&A.

What is structured output prompting?

31

Answer:

- Instructing the model to respond in JSON, XML, or another strict format.
- Combined with function calling for reliable, parseable AI outputs.
- Essential for production pipelines where downstream code parses results.

What is a prompt template?

32

Answer:

- A reusable prompt structure with variables filled at runtime.
- Standardizes LLM inputs across a pipeline — improves consistency.
- Frameworks like LangChain and LlamaIndex provide template management.

What is automatic prompt optimization (APO)?

33

Answer:

- Using an optimizer (gradient-based or LLM-based) to improve prompts.
- DSPy compiles high-level task specs into optimized prompt programs.
- Replaces manual prompt iteration — critical for complex multi-step tasks.

What is context window management in production?

34

Answer:

- Selecting and prioritizing what fits within token limits for each request.
- Strategies: summarization, sliding window, retrieval, priority scoring.
- Poor context management is a top cause of production LLM failures.

What is prompt chaining?

35

Answer:

- Breaking a complex task into sequential prompts where output feeds next.
- Each step is simpler and more reliable than one mega-prompt.
- Enables auditing, debugging, and human review at each chain step.

What is step-back prompting?

36

Answer:

- Ask the model a higher-level abstract question before the specific one.
- Retrieves relevant background knowledge before reasoning on the task.
- Improves accuracy on physics, history, and complex reasoning benchmarks.

What is Tree-of-Thought (ToT) prompting?

37

Answer:

- Explores multiple reasoning paths as a tree — evaluates and prunes branches.
- More powerful than linear CoT for tasks requiring search or planning.
- Requires more inference calls but solves harder problems reliably.

How does RAG work end-to-end?

41

Answer:

- Offline: chunk docs → embed → store in vector DB with metadata.
- Online: embed query → ANN search → inject top-k chunks into prompt.
- LLM generates grounded answer; response cites retrieved sources.

What is the difference between dense and sparse retrieval?

42

Answer:

- Sparse (BM25): exact lexical match — great for specific terms/IDs, fast.
- Dense (DPR, E5): neural embeddings — handles paraphrase and synonyms.
- Hybrid retrieval + RRF fusion beats either alone in most benchmarks.

What is a vector database?

43

Answer:

- Stores high-dimensional embeddings and supports ANN (approximate nearest-neighbor) search.
- Examples: Pinecone, Weaviate, Qdrant, Chroma, pgvector (PostgreSQL).
- Core infrastructure component for any RAG or semantic search system.

What is chunking strategy in RAG and why does it matter?

44

Answer:

- How documents are split: fixed-size, sentence-based, or semantic chunking.
- Poor chunking breaks context across chunks — degrades retrieval quality.
- Overlap between chunks prevents losing information at split boundaries.

What is reranking in RAG pipelines?

45

Answer:

- Cross-encoder model reorders retrieved chunks by relevance to query.
- First-stage retrieval is fast (ANN); reranker is slower but more accurate.
- Significantly reduces hallucination by ensuring top chunks are truly relevant.

What are the trade-offs between fine-tuning and RAG?

46

Answer:

- Fine-tuning: knowledge baked into weights — fast inference, no retrieval lag.
- RAG: knowledge stays external and updatable — no costly retraining.
- Hybrid (fine-tune + RAG) is the dominant production architecture.

What is HyDE (Hypothetical Document Embeddings)?

47

Answer:

- Generate a hypothetical answer to the query, embed it for retrieval.
- The hypothetical doc is closer to real docs than the raw query embedding.

What is multi-vector retrieval?

51

Answer:

- Store multiple embeddings per document (e.g., summary + full doc + propositions).
- ColBERT: store token-level embeddings for late interaction retrieval.
- Improves recall for complex queries that dense bi-encoders miss.

What is query transformation in RAG?

52

Answer:

- Rewriting, expanding, or decomposing the user query before retrieval.
- Multi-query: generate N query variants, retrieve for each, merge results.
- Reduces vocabulary mismatch between user queries and indexed documents.

What is metadata filtering in RAG?

53

Answer:

- Filter retrieved docs by structured fields (date, author, category) before ranking.
- Combines ANN search with SQL-like constraints for precise retrieval.
- Essential for enterprise RAG where access control per-doc is required.

What is iterative / agentic RAG?

54

Answer:

- Agent retrieves, reads results, decides if more retrieval is needed, loops.
- Handles multi-hop questions requiring reasoning across multiple documents.
- Frameworks: LangGraph, LlamaIndex Agents — orchestrate retrieval loops.

What is embedding model selection for RAG?

55

Answer:

- Choose based on MTEB benchmark scores for your task and language.
- Top models: text-embedding-3-large (OpenAI), E5-Mistral, BGE-M3.
- Domain-specific fine-tuning of embeddings further boosts retrieval recall.

What is faithfulness vs. relevance in RAG evaluation?

56

Answer:

- Faithfulness: is the answer grounded in retrieved context — no hallucination?
- Relevance: are retrieved chunks actually relevant to the query?
- RAGAS framework measures both automatically for RAG pipeline evaluation.

What is the document ingestion pipeline for RAG?

57

Answer:

- Load → parse (PDF, HTML, DOCX) → clean → chunk → embed → index → store.
- Handle tables, images, and code blocks separately for better extraction.
- Unstructured.io, LlamaParse, and Docling are popular parsing libraries.

What is an AI agent?

61

Answer:

- LLM used as reasoning engine in a loop: perceive → think → act → observe.
- Unlike single LLM call, agents handle multi-step tasks and use tools.
- Frameworks: LangGraph, AutoGen, CrewAI, OpenAI Assistants API.

What is the Model Context Protocol (MCP)?

62

Answer:

- Open standard by Anthropic for connecting AI to external tools/data sources.
- Defines a standard interface so any MCP tool plugs into any MCP host.
- Analogy: USB for AI — replaces bespoke integrations with universal connectors.

What is tool use / function calling in agents?

63

Answer:

- Agent calls external tools: web search, code execution, DB queries, APIs.
- LLM outputs structured tool call; host executes it and returns result.
- Enables agents to act in the real world beyond text generation.

What are multi-agent systems?

64

Answer:

- Specialized agents (planner, retriever, coder, critic) collaborate via messages.
- Orchestrator delegates subtasks; agents return results to the pipeline.
- Frameworks: AutoGen, CrewAI, MetaGPT for building multi-agent workflows.

What are the main failure modes of AI agents?

65

Answer:

- Error propagation: one agent's mistake cascades through the pipeline.
- Infinite loops: agents ping-pong without progress — need timeout logic.
- Prompt injection via environment: malicious content hijacks agent actions.

What is the difference between reactive and planning agents?

66

Answer:

- Reactive: respond to immediate observations without long-term planning.
- Planning (Plan-and-Execute): generate a plan first, then execute step by step.
- Planning agents handle complex multi-step goals more reliably.

What is memory in AI agents?

67

Answer:

- Short-term: conversation history in the context window for current session.
- Long-term: persistent store (vector DB, DB) recalled across sessions.
- Procedural: learned behaviors/skills stored for reuse across tasks.

What is an agent scratchpad?

71

Answer:

- Intermediate working memory where the agent records its reasoning steps.
- ReAct agents store thought → action → observation in the scratchpad.
- Provides transparency and enables debugging of agent decision chains.

What is a tool manifest / tool schema?

72

Answer:

- JSON schema describing available tools: name, description, parameters, types.
- LLM uses the schema to decide when and how to call each tool correctly.
- Good tool descriptions are as important as the tools themselves.

What is code execution as a tool?

73

Answer:

- Agent writes code, executes it in a sandbox, reads output, iterates.
- Enables precise computation, data analysis, and file manipulation.
- OpenAI Code Interpreter and Claude Code both implement this pattern.

What is the difference between agents and workflows?

74

Answer:

- Workflows: fixed sequence of LLM + tool calls — deterministic pipelines.
- Agents: dynamic — LLM decides the next step based on current context.
- Use workflows when the process is known; agents when it needs to adapt.

What is LangGraph and why is it popular for agents?

75

Answer:

- Graph-based framework for stateful, multi-actor agentic applications.
- Nodes = processing steps; edges = transitions — explicit control flow.
- Supports cycles, branching, and human-in-the-loop checkpoints natively.

What is an orchestrator vs sub-agent?

76

Answer:

- Orchestrator: high-level planner that delegates tasks to specialized sub-agents.
- Sub-agents: specialized workers — each focused on one capability domain.
- Separation of concerns makes complex agentic systems more maintainable.

What is agent evaluation and how is it done?

77

Answer:

- Evaluate task completion rate, tool call accuracy, and trajectory efficiency.
- Use LLM-as-judge to score intermediate steps, not just final output.
- Build eval datasets from real user sessions — test regressions on every release.

What is LoRA (Low-Rank Adaptation)?

81

Answer:

- Adds low-rank trainable matrices $B \cdot A$ to frozen weight matrices W .
- Only B and A are trained — reduces parameters from millions to thousands.
- $r=8-16$ typically captures enough task-relevant variation; QLoRA adds 4-bit.

What is PEFT (Parameter-Efficient Fine-Tuning)?

82

Answer:

- Updates only a small subset of parameters while freezing the rest.
- Variants: LoRA, prefix tuning, prompt tuning, IA³, adapter layers.
- Enables multi-task serving via adapter swapping on a single base model.

What is DPO (Direct Preference Optimization)?

83

Answer:

- Re-frames RLHF as a single supervised objective — no reward model needed.
- Given preference pairs (chosen, rejected), trains with binary cross-entropy.
- Simpler than PPO-based RLHF, more stable training, competitive results.

What is catastrophic forgetting in LLM fine-tuning?

84

Answer:

- Fine-tuning overwrites pretrained knowledge — model forgets prior abilities.
- Mitigate: LoRA/PEFT (freeze base), replay training data mix, small LR.
- EWC penalizes changes to weights important for previous tasks.

What is QLoRA?

85

Answer:

- Quantizes base model to 4-bit; applies LoRA adapters in higher precision.
- Enables fine-tuning 65B+ models on a single 48GB GPU — massive cost saving.
- Near full fine-tuning quality at a fraction of GPU memory requirement.

What is the pretraining pipeline of an LLM?

86

Answer:

- Data collect → clean/dedup → tokenizer train → distributed training → eval.
- Runs for trillions of tokens over weeks on thousands of GPUs.
- Scaling laws (Chinchilla): optimal $N^* \approx \text{training tokens} / 20$.

What are Chinchilla scaling laws?

87

Answer:

- Given compute budget C , optimal model size $N^* \propto \sqrt{C}$, tokens $D^* \propto \sqrt{C}$.
- Chinchilla-optimal: 70B model trained on 1.4T tokens (not just more params).
- Llama philosophy: smaller model on more tokens = better inference efficiency.

What is SFT (Supervised Fine-Tuning)?

91

Answer:

- First phase of alignment: fine-tune base LLM on curated demonstrations.
- Teaches format, dialogue structure, and basic instruction following.
- Foundation before RLHF — without good SFT, reward model training fails.

What is the PPO algorithm in RLHF?

92

Answer:

- Proximal Policy Optimization: optimizes policy with clipped probability ratio.
- KL penalty $\lambda \cdot \text{KL}(\pi_\theta || \pi_{\text{SFT}})$ prevents over-optimization away from base model.
- Challenges: reward hacking, mode collapse, KL-divergence calibration.

What is adapter merging and Task Arithmetic?

93

Answer:

- Task vector $\tau = \theta_{\text{fine_tuned}} - \theta_{\text{base}}$; merge by adding task vectors.
- Enables zero-cost multi-task deployment without joint retraining.
- TIES merging resolves sign conflicts among task vectors for better quality.

What is continual learning in LLMs?

94

Answer:

- Training on sequential tasks/domains without forgetting previous knowledge.
- Key challenge: catastrophic forgetting in standard gradient descent.
- Approaches: replay buffers, EWC regularization, modular adapter systems.

What is data curation for LLM fine-tuning?

95

Answer:

- Quality > quantity: 1k high-quality examples often beats 100k noisy ones.
- Dedup, filter, balance — remove redundant, toxic, or low-signal data.
- Alpaca, ShareGPT, UltraChat are common open instruction-tuning datasets.

What is the role of learning rate in fine-tuning?

96

Answer:

- Very small LR ($1e-5$ to $5e-5$) preserves pretrained knowledge while adapting.
- Too high: catastrophic forgetting. Too low: no adaptation from fine-tuning.
- Warmup + cosine decay schedule is standard for transformer fine-tuning.

What is multi-task fine-tuning?

97

Answer:

- Simultaneously fine-tune on multiple task datasets — improves generalization.
- Prevents overfitting to a single task; improves instruction following breadth.
- T5 and Flan-T5 pioneered this; Flan collection has 1800+ fine-tuning tasks.

What is KV caching in LLM inference?101
Answer:

- Stores key/value tensors from previous tokens — avoids recomputing them.
- Reduces per-step compute from $O(n)$ to $O(1)$ for cached prefix positions.
- Memory grows linearly with sequence length and batch size.

What is PagedAttention?102
Answer:

- Manages KV cache like OS virtual memory using fixed-size pages/blocks.
- Near-zero fragmentation, memory sharing for parallel sampling/beam search.
- Core of vLLM — enables 2-4× throughput improvement over naive serving.

What is continuous batching in LLM serving?103
Answer:

- New requests join the batch mid-flight when other sequences finish.
- Eliminates idle GPU time waiting for entire batch to complete together.
- Combined with PagedAttention, dramatically increases GPU utilization.

What is model quantization?104
Answer:

- Reduces weight precision FP32→INT8/INT4 — smaller footprint, faster inference.
- GPTQ: one-shot PTQ via Hessian-based error minimization, near-FP16 quality.
- AWQ preserves salient weights; enables 70B models on single 48GB GPU.

What are TTFT and TPS metrics in LLM serving?105
Answer:

- TTFT (time-to-first-token): dominated by prefill phase — latency metric.
- TPS (tokens-per-second): dominated by decode phase — throughput metric.
- Disaggregating prefill and decode across GPU pools optimizes both independently.

What is speculative decoding?106
Answer:

- Small draft model proposes k tokens; large target verifies all in one pass.
- Accepted tokens kept; rejected token corrected — output distribution unchanged.
- 2-4× speedup with zero quality degradation — ideal for long outputs.

What is tensor parallelism vs pipeline parallelism?107
Answer:

- Tensor: split individual weight matrices across GPUs — all-reduce each layer.
- Pipeline: partition layers across devices — send activations between stages.
- 3D parallelism (tensor + pipeline + data) trains models beyond single node.

What is semantic caching for LLM APIs?111
Answer:

- Cache LLM responses; for semantically similar queries, return cached result.
- GPTCache embeds queries and retrieves cached responses by similarity.
- Reduces costs dramatically for high-traffic applications with repeated questions.

What is a model gateway / LLM router?112
Answer:

- Middleware intercepting API calls — routes to appropriate model/backend.
- Handles: auth, rate limiting, cost tracking, logging, provider failover.
- Examples: LiteLLM, Portkey, AWS Bedrock gateway, Kong AI Gateway.

What is vLLM and why is it the industry standard?113
Answer:

- High-throughput LLM serving library with PagedAttention + continuous batching.
- Supports most open-source models; OpenAI-compatible API out of the box.
- Used in production at major AI labs and cloud providers for serving.

What is mixed precision training?114
Answer:

- Use BF16/FP16 for forward/backward pass; FP32 master weights for updates.
- Halves GPU memory and speeds up compute without significant accuracy loss.
- BF16 preferred over FP16 for training — wider dynamic range, less overflow.

What is weight tying in language models?115
Answer:

- Input embedding weights shared with output projection (lm_head) weights.
- Reduces parameters significantly for vocabulary-heavy models.
- Forces embedding and unembedding to be in the same semantic space.

What is the prefill vs decode phase in LLM inference?116
Answer:

- Prefill: process entire input prompt in parallel — GPU compute-bound.
- Decode: generate output tokens one by one — GPU memory-bandwidth-bound.
- Different hardware and optimization strategies needed for each phase.

What is dynamic batching in LLM serving?117
Answer:

- Group requests with similar sequence lengths to minimize padding waste.
- Padding different-length sequences wastes compute on padding tokens.
- Variable-length batching with careful scheduling maximizes GPU utilization.

What is n8n and how is it used with AI?

121

Answer:

- Open-source visual automation tool — connects any app or API via nodes.
- Self-host for free; 400+ integrations including Claude, OpenAI, Google.
- Build AI workflows: trigger → enrich with LLM → store/notify/act.

What is Zapier AI and what are Zaps?

122

Answer:

- No-code automation: Trigger → Action across 6,000+ apps without coding.
- AI Zaps add LLM steps: summarize, classify, extract, translate inline.
- Best for non-developers; more limited than n8n for complex branching logic.

What is Make (formerly Integromat)?

123

Answer:

- Visual automation builder with powerful data mapping and error handling.
- More flexible than Zapier for complex multi-branch workflows.
- Strong AI integrations: OpenAI, Claude, Gemini modules available natively.

What is LangChain?

124

Answer:

- Python/JS framework for building LLM applications with chains and agents.
- Provides: prompt templates, memory, tools, retrievers, agent executors.
- LCEL (LangChain Expression Language) composes chains declaratively.

What is LlamaIndex?

125

Answer:

- Data framework for connecting LLMs to structured and unstructured data.
- Best-in-class for RAG: ingestion, chunking, indexing, retrieval pipelines.
- Supports multi-modal, knowledge graphs, and agentic retrieval patterns.

What is DSPy?

126

Answer:

- Framework that compiles high-level task descriptions into optimized prompts.
- Uses automatic prompt optimization — replaces manual prompt engineering.
- Teleprompter finds best few-shot examples and instructions automatically.

What is Zapier's NLA (Natural Language Actions)?

127

Answer:

- Exposes Zapier actions as LLM-callable tools via natural language.
- Lets AI agents trigger any Zapier-connected app through conversational commands.
- Powers AI assistants that can book calendars, send emails, update CRMs.

What is the OpenAI Assistants API?

131

Answer:

- Managed agent infrastructure: threads, messages, tools, file search.
- Handles state management and tool execution automatically.
- Good for rapid prototyping; less control than building custom agent loops.

What is Crew AI?

132

Answer:

- Multi-agent framework focused on role-based agent collaboration.
- Agents have roles, goals, and backstories — collaborate on complex tasks.
- Simpler API than LangGraph — good for content creation and research tasks.

What is a webhook in AI automation?

133

Answer:

- HTTP endpoint that receives real-time event notifications from services.
- Triggers automation workflows when events occur: new email, form submit, etc.
- Core building block for real-time AI pipelines: event → enrich → act.

What is Flowise?

134

Answer:

- Visual drag-and-drop builder for LangChain/LlamaIndex-based AI apps.
- Self-hostable; builds chatbots, RAG pipelines, and agents visually.
- Exports to code — good bridge between no-code prototyping and production.

What is Botpress?

135

Answer:

- Conversational AI platform for building production chatbots and agents.
- Visual flow editor + LLM integration + analytics + channel deployment.
- Used by enterprises to deploy AI assistants across web, WhatsApp, Slack.

What is Relevance AI?

136

Answer:

- No-code platform for building AI agents and tools without engineering.
- Connects to any API, data source, or LLM with a visual builder interface.
- Popular for sales automation, customer support, and research workflows.

What is a trigger-action paradigm in AI automation?

137

Answer:

- Every automation starts with a trigger (event) and executes actions.
- AI adds intelligence: classify trigger → choose action → personalize output.
- Pattern used by Zapier, n8n, Make, and every major automation platform.

What is the bias-variance trade-off?

141

Answer:

- Bias: error from wrong assumptions — underfitting. Variance: overfitting.
- As model complexity grows, bias falls and variance rises — sweet spot needed.
- Ensemble methods: bagging reduces variance; boosting reduces bias.

What is gradient descent and its variants?

142

Answer:

- $\theta \leftarrow \theta - \alpha \cdot \nabla L$ — update weights in direction of negative gradient.
- SGD (1 sample), mini-batch (B samples), Adam (adaptive per-param LR).
- Adam uses first and second moment estimates — fast convergence, less LR tuning.

What is regularization (L1 vs L2)?

143

Answer:

- L2 (Ridge): $\lambda \|w\|^2$ penalty — shrinks weights proportionally, never zero.
- L1 (Lasso): $\lambda \|w\|$ penalty — constant shrinkage, produces sparse weights.
- L1 induces feature selection; L2 distributes weight across all features.

What is cross-validation?

144

Answer:

- k-fold: split data k ways; train on k-1, validate on 1, rotate, average.
- Gives more reliable performance estimate than single train/val split.
- Stratified CV preserves class balance — essential for imbalanced datasets.

What is XGBoost and why is it powerful?

145

Answer:

- Gradient boosting with second-order Taylor expansion of loss (uses Hessian).
- Adds L1/L2 regularization on tree structure — prevents overfitting.
- Sparse-aware split finding, cache-aware computation — extremely fast.

What is the ROC-AUC metric?

146

Answer:

- ROC plots TPR vs FPR at all thresholds; AUC = area under curve.
- AUC=1 perfect, AUC=0.5 random — threshold-independent performance measure.
- On imbalanced data, prefer PR-AUC which focuses on minority class.

What is SHAP and how does it explain model predictions?

147

Answer:

- Assigns each feature a Shapley value: average marginal contribution.
- Satisfies efficiency, symmetry, dummy, and additivity axioms.
- TreeSHAP computes exact values for tree models in polynomial time.

What is PCA (Principal Component Analysis)?

151

Answer:

- Linear dimensionality reduction finding axes of maximum variance.
- Computed via SVD of data matrix — reduces dimensions preserving most variance.
- Used for visualization, noise reduction, and preprocessing before ML.

What is the curse of dimensionality?

152

Answer:

- As features grow, data becomes sparse — volume grows exponentially.
- Distance metrics lose meaning; nearest-neighbor search degrades rapidly.
- Mitigate: PCA, feature selection, regularization, or learned representations.

What is precision vs recall?

153

Answer:

- Precision = $TP / (TP + FP)$: how many predicted positives are actually positive.
- Recall = $TP / (TP + FN)$: how many actual positives are correctly identified.
- Prefer precision when FP costly (spam); recall when FN costly (cancer screening).

What is class imbalance and how is it handled?

154

Answer:

- When one class has far more samples — model biased toward majority class.
- Oversampling (SMOTE), undersampling, class weights in loss, or PR-AUC metric.
- Ensemble methods often more robust to imbalance than single models.

What is early stopping?

155

Answer:

- Monitor validation loss; stop training when it stops improving for p epochs.
- Acts as implicit regularization — limits effective model capacity.
- Save checkpoint at best validation loss — not at training end.

What is a confusion matrix?

156

Answer:

- Grid of actual vs. predicted class counts — rows actual, columns predicted.
- Diagonal = correct predictions; off-diagonal = errors by type.
- Derives precision, recall, F1, specificity — complete error analysis.

What is ensemble learning?

157

Answer:

- Combine multiple models — errors uncorrelated, averaging cancels them.
- Bagging (Random Forest): parallel training, reduces variance.
- Boosting (XGBoost): sequential, focuses on previous errors, reduces bias.

What is backpropagation?

161

Answer:

- Applies chain rule to compute gradients of loss w.r.t. every parameter.
- Gradients flow backward from output to input using cached forward activations.
- Foundation of training all neural networks — automatic differentiation.

What is residual connection (ResNet)?

162

Answer:

- Skip connection: $\text{output} = F(x) + x$ — gradient flows through skip path.
- Enables training 100+ layer networks by alleviating vanishing gradients.
- Also enables learning residuals (small corrections) rather than full mappings.

What is layer normalization?

163

Answer:

- Normalizes across feature dimension per sample — batch-size independent.
- Preferred over BatchNorm for sequence models and transformers.
- RMSNorm (LLaMA) removes mean centering — faster with similar performance.

What is the Vision Transformer (ViT)?

164

Answer:

- Divides image into patches; each patch embedded as a token for Transformer.
- Class token prepended; positional embeddings added to patch tokens.
- Competitive with CNNs at scale — DINOv2 shows excellent transfer learning.

What is self-supervised learning (SSL)?

165

Answer:

- Trains representations without labels using data itself as supervision.
- Contrastive: SimCLR/MoCo contrast augmented views of the same image.
- Masked: MAE masks patches and reconstructs pixels — learns rich features.

What is knowledge distillation in deep learning?

166

Answer:

- Train small student to match large teacher's soft probability outputs.
- Soft targets at high temperature reveal inter-class similarity — better learning.
- DistilBERT: 40% smaller, 60% faster, retains 97% of BERT performance.

What is GELU activation and why is it used in transformers?

167

Answer:

- $\text{GELU}(x) = x \cdot \Phi(x)$ — smooth, non-zero gradient everywhere unlike ReLU.
- Slight negative output for small negative x — better gradient flow.
- Used in BERT, GPT, and virtually all modern transformer architectures.

What is contrastive learning?

171

Answer:

- Train encoder to embed similar pairs close, dissimilar pairs far apart.
- SimCLR: two augmented views of same image should be closer than other images.
- NT-Xent loss; large batch provides many negatives — critical for quality.

What is the encoder-decoder architecture?

172

Answer:

- Encoder: bidirectional attention — reads and represents source sequence.
- Decoder: causal attention + cross-attention to encoder — generates target.
- Cross-attention allows decoder to attend to relevant encoder positions.

What is dropout and how does it prevent overfitting?

173

Answer:

- Randomly zeros neuron activations with probability p during training.
- Forces redundant representations — no neuron can rely on any other neuron.
- At inference: all neurons active, outputs scaled by $(1-p)$. Ensemble effect.

What is weight initialization and why does it matter?

174

Answer:

- Poor init $\rightarrow$ vanishing/exploding gradients from the very start of training.
- Xavier: $\text{Var}(w) = 2/(\text{fan_in} + \text{fan_out})$ — good for tanh/sigmoid activations.
- He: $\text{Var}(w) = 2/\text{fan_in}$ — optimized for ReLU; default in modern deep learning.

What is the Transformer architecture overview?

175

Answer:

- Multi-head self-attention $\rightarrow$ Add & LayerNorm $\rightarrow$ FFN $\rightarrow$ Add & LayerNorm per block.
- Decoder adds cross-attention between self-attention and FFN layers.
- Scales remarkably well: more data + more compute = better performance.

What is a VAE (Variational Autoencoder)?

176

Answer:

- Encoder outputs mean and variance of Gaussian posterior $q(z|x)$.
- Reparameterization: $z = \mu + \sigma \cdot \epsilon$ enables backprop through sampling.
- Loss = reconstruction + KL divergence — forces structured latent space.

How do you evaluate an LLM for a production use case?

181

Answer:

- Define task-specific metrics: accuracy, faithfulness, latency, refusal rate.
- Layer evals: automated benchmarks → LLM-as-judge → human gold-standard.
- CI/CD gating: run eval suite on every model or prompt change before deploy.

What is red-teaming for AI models?

182

Answer:

- Adversarial testing: attempt to make the model produce harmful outputs.
- Covers: jailbreaks, prompt injection, bias elicitation, privacy leakage.
- Anthropic, OpenAI use external red teams + automated red-teaming tools.

What is an AI product tech stack for 2025?

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Answer:

- LLM layer: Claude/GPT-4o. Orchestration: LangGraph/LlamaIndex.
- Memory: Postgres+pgvector, Pinecone. Serving: vLLM/Modal/AWS Bedrock.
- Eval: RAGAS, LangSmith. Observability: Langfuse, Helicone, Braintrust.

What is responsible AI and why does it matter?

184

Answer:

- Ensures AI systems are safe, fair, transparent, and accountable.
- Covers: bias, privacy, security, environmental impact, human oversight.
- Regulatory frameworks (EU AI Act) increasingly mandate responsible AI.

What is AI observability?

185

Answer:

- Monitor LLM inputs, outputs, latency, cost, and errors in production.
- Tools: Langfuse, Helicone, Braintrust, W&B Weave, LangSmith.
- Essential for debugging hallucinations and catching prompt regressions.

What is the EU AI Act and how does it affect AI engineers?

186

Answer:

- Risk-based regulation: unacceptable, high, limited, minimal risk categories.
- High-risk AI (hiring, credit, healthcare) requires documentation and auditing.
- General Purpose AI (GPAI) models like GPT-4/Claude face transparency requirements.

What is AI for code review and how does it work?

187

Answer:

- LLM analyzes PR diffs: finds bugs, style violations, security issues.
- GitHub Copilot, CodeRabbit, Graphite Automations integrate into CI/CD.
- Best used for: catching obvious issues; humans still needed for architecture

What is an AI side hustle and what's realistic?

191

Answer:

- AI Freelancing: automation, chatbots, RAG systems on Upwork/Fiverr.
- Content: AI-powered newsletters, YouTube tutorials, prompt packs on Gumroad.
- Results depend entirely on skill, consistency, and niche specificity.

What is the AI engineering career path?

192

Answer:

- Entry: prompt engineer / AI assistant developer — build on top of APIs.
- Mid: AI engineer — RAG systems, fine-tuning, agent pipelines, evals.
- Senior: ML engineer / AI researcher — training, architecture, infrastructure.

What skills does an AI engineer need in 2025?

193

Answer:

- Prompt engineering, Python, LLM APIs (Claude, OpenAI, Gemini).
- RAG pipelines, vector databases, agent frameworks (LangGraph, LlamaIndex).
- ML fundamentals, fine-tuning (LoRA), evaluation, and production deployment.

What is AI for interview preparation?

194

Answer:

- Use Claude to roleplay technical and behavioral interviews with feedback.
- Generate domain-specific question banks; practice explaining concepts simply.
- Research companies, industries, and role requirements with AI assistance.

What is a Claude Project and how do AI creators use it?

195

Answer:

- Persistent workspace with uploaded docs and custom system instructions.
- Creators upload brand guidelines, past content, audience persona — AI stays aligned.
- Eliminates repeated context-setting — enables consistent content at scale.

What is AI content automation for creators?

196

Answer:

- Repurpose long-form content: video → transcript → blog → tweets → carousel.
- Tools: Descript (video), Castmagic (podcast), Claude (text transformation).
- AI handles production; human handles strategy, voice, and authentic connection.

What is multimodal AI and its applications?

197

Answer:

- AI that processes text, images, audio, video together in one model.
- Applications: document analysis, image captioning, video understanding.
- GPT-4o, Claude 3.5, Gemini 1.5 Pro — all multimodal frontier models